

MAMA-SAUTI — ASR Benchmark Report

Intron Sahara CodeSwitch Africa Challenge · Health Track | Generated 2026-09-13 | github.com/bellahellam/Intron_Sahara_Codeswitch_Africa_Challenge

1. Purpose

MAMA-SAUTI is a postpartum mental health screening assistant that listens to a Kenyan mother speak in natural Kiswahili-English code-switching and produces a PHQ-9/GAD-7 screening record. The benchmark asks one question: does ASR quality on code-switched Swahili **change the clinical decision** this product makes?

The dominant failure on this type of speech is **switch-boundary deletion** — the model returns fluent monolingual Swahili with the embedded English gone. In this setting that English carries the affective vocabulary ("stress", "depressed", "I can't cope"). Aggregate WER cannot distinguish losing a filler from losing the only clinical content in a sentence. This benchmark adds two metrics WER cannot see: **EESR** (Embedded-English Span Recall) and **EESR-clinical** (restricted to spans containing an affective term).

2. Data

Property	Value
Dataset	intronhealth/AfriSwitchCare, config <i>swahili</i> , split <i>test</i> (CC BY-NC-SA 4.0, gated)
Language	Kiswahili-English intra-utterance code-switched
Size	12 simulated doctor-patient consultations · 1.54 hours · mean CMI 37.4 (highest switch density of 8 languages in corpus)
Chunking	Fixed 75 s windows, 1 s overlap — computed once, shared byte-identically across all models to void segmentation as a confound
Preprocessing	Failed chunks recorded as failures (not empty strings — an empty transcript scores as perfect deletion). WER/CER: Intron normalisation (diacritics removed). EESR/CIR/SPR: diacritic-preserving normalisation (diacritic stripping collapses lexicon-matching distinctions).

3. Models

Model	Type	Notes
Sahara v2.5 (corrections OFF)	Cloud API	Shipped configuration. Corrections OFF: every score is justified by a verbatim quote read back to the patient — a smoothed rewrite is not her words. Primary column.
ElevenLabs Scribe v2	Cloud API	Strong multilingual model; no benchmarking ToS restriction. ~\$0.22/hr.
Whisper large-v3 (auto-detect)	Local · CPU · int8	Language auto-detected per chunk via faster-whisper.
Whisper large-v3 (sw forced)	Local · CPU · int8	Language forced to <i>sw</i> . Tests whether the most-used open ASR model collapses code-switching when instructed to treat audio as monolingual Swahili.
Jacaranda-Health/ASR-STT	Local · CPU	Whisper-medium fine-tuned on Swahili+English by the organisation running maternal-health messaging for ~3M Kenyan mothers. The regional incumbent.

Excluded (with reasons): Deepgram (ToS \$9 prohibits competitive benchmarking, unconditionally) · AssemblyAI (prohibits competitive analysis and third-party corpus evaluation) · NVIDIA Parakeet/Canary (no Swahili checkpoint) · w2v-BERT 2.0 (bare checkpoint, requires fine-tuning, cannot benchmark zero-shot).

4. Metrics

Metric	↓/ ↑	Rationale
WER	↓	Standard. Reported under Intron's normalisation for comparability with published baselines.
CER	↓	More robust than WER for morphologically rich languages where a token error affects multiple characters.

EESR	↑	For each gold [[EN]]-tagged span, did its token sequence survive? Directly measures switch-boundary deletion. Net-new metric absent from Intron's benchmarking repo.
EESR-clinical	↑	EESR restricted to spans containing an affective term. A model can have healthy EESR while this collapses — and this is the subset that determines whether screening works.
CMI-Δ	↓	Difference in Code-Mixing Index between reference and hypothesis. Measures whether the model flattens the utterance's switching structure.
p95 latency/chunk	↓	Deployment constraint: product targets a mid-range Android in a low-connectivity clinic.

CIR and SPR are computed but shown as raw counts only (n=1 in this corpus — a rate over n=1 is not a rate). Tier 3 (band-flip, | ΔPHQ-9|) is absent by design: 11/12 conversations are non-psychiatric with PHQ-9 totals near zero; no transcription error can move the band. Stated explicitly, not hidden.

5. Quantitative Results

12 conversations · 1.54 h · Kiswahili-English code-switched · AfriSwitchCare Swahili test

Model	WER ↓	CER ↓	EESR ↑	EESR-cl ↑	CMI-Δ ↓	p95 ms/chunk
ElevenLabs Scribe v2	0.171	0.075	89.1%	80.0%	4.65	7,253
Sahara v2.5 (shipped, corr. OFF) ★	0.194	0.107	77.3%	73.3%	4.92	7,389
Whisper v3 (auto-detect)	0.332	0.149	82.5%	93.3%	3.37	42,894
Whisper v3 (sw forced)	0.289	0.138	78.7%	— †	10.32	149,686
Jacaranda-Health/ASR-STT	0.565	0.393	35.5%	20.0%	13.16	60,890

★ Deployed model. EESR computed over 616 embedded-English spans; EESR-clinical over the 15 containing an affective term.
† — means the model deleted *all* embedded English on clinical-affective spans — zero survived to score. This is the finding, not missing data.

Published baselines (context)	
shamiriAI (Lilan et al. 2026, JMIR AI 5:e95063) — code-switched Kiswahili/English/Sheng mental health audio	WER 0.34, CER 0.19 ← realistic ceiling
AfriSwitch best system, code-switched Swahili	WER 0.341
Intron MultiBench — monolingual accented clinical Swahili, Sahara	WER 0.068 ← <i>do not compare; gap to 0.34 is the competition's thesis</i>

6. Qualitative Findings

ElevenLabs Scribe v2

Strengths: Best WER and CER. Strong English span retention (EESR 89.1%). Low latency. Handles code-switching without language forcing.

Weaknesses: EESR-clinical (80%) lower than Whisper auto despite better WER — some affective terms transcribed in paraphrase. Cloud-only; data leaves device (consent implications). Cost at scale.

Sahara v2.5 (shipped, corrections OFF)

Strengths: Lowest latency (p95 7.4 s/chunk). Competitive WER for an African-language model. No switch-boundary catastrophe.

Weaknesses: EESR-clinical 73.3% — lowest among models with a score. Corrections deliberately disabled; corrections-ON would score differently but introduces an undisclosed post-processor that makes the safety scan non-deterministic.

Whisper v3 (auto-detect)

Whisper v3 (sw forced)

Strengths: Lower aggregate WER than auto-detect (0.289). Faster than auto-detect.

Weaknesses: The critical finding of this benchmark. Forcing *language="sw"* causes the model to delete all embedded English. EESR-clinical is unmeasurable — zero affective English spans survived across all 12 conversations. CMI-Δ 10.32 confirms the switching structure is flattened. A lower WER while losing all clinical content is the exact failure mode this benchmark was built to detect. This configuration is unsafe for this use case.

Jacaranda-Health/ASR-STT

Strengths: Regional incumbent with direct maternal-health deployment experience. No API cost. Relevant baseline: if Sahara did not beat this model, that would be a publishable negative finding.

Weaknesses: Worst on every metric — WER 0.565,

Strengths: Best EESR-clinical (93.3%) — preserves affective English better than any other model. Best CMI- Δ (3.37). No API cost.

Weaknesses: Higher WER (0.332). p95 latency 43 s/chunk on CPU — not viable for real-time use without GPU.

EESR 35.5%, EESR-clinical 20.0%. Loses the majority of embedded English and scores near-zero on affective terms for most conversations.

7. Key Finding

The question is not WER — it is what survives at the switch boundary.

Whisper sw-forced achieves lower WER than auto-detect but loses *all* clinical English. Jacaranda, the regional incumbent, loses most of it. Sahara achieves competitive WER with the lowest latency but lower EESR-clinical than either Whisper condition. ElevenLabs leads on standard metrics. For a product where the clinical decision depends on affective English vocabulary surviving transcription, **EESR-clinical is the metric that matters.**

Sahara v2.5 (corrections OFF) is the deployed configuration because it is the only model that combines: (a) sub-10 s chunk latency for real-time use in a low-connectivity clinic, (b) no switch-boundary deletion catastrophe, and (c) a deterministic transcript that can be quoted verbatim back to the patient.

Every number is auditable to a hypothesis transcript. Reproduce: `python -m bench.run && python -m bench.report`.